

EE 554 : Low Power VLSI Design

Course Code : EE 554

Course Name : Low Power VLSI Design

L-T-P-C : 3-0-0-3

Intended for : B. Tech in Engineering (EE), Microelectronics and VLSI, M. Tech in Microelectronics and VLSI Design, MS, M. Tech. & PhD

Prerequisite : Digital System Design (EE210)

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction to low power design:** Components of power dissipation of VLSI circuits. (2 Hours)
- **Circuit techniques for Low Power Design:** Standby leakage control using transistor stacks, multi-threshold and dynamic threshold techniques, supply voltage scaling technique. (4 Hours)
- **Low power low voltage arithmetic circuits:** Low power adder architectures – ripple carry adder, carry look-ahead adder, carry select adder, carry save adder, carry skip adder, current mode adder using multi-valued logic, residue adders, low power multiplier architectures- serial multiplier, parallel multiplier, serial-parallel multiplier, Braun multiplier, Baugh Wooley multiplier, Booth multiplier, Wallace tree multiplier. (10 Hours)
- **Low power low voltage memories:** Read only and random access memories. Power reduction of read only and random access memories at architectural, logic, and transistor levels. (8 Hours)
- **Transforms for low power VLSI circuits:** Behavioral level transform, algorithm and architecture level transform, negative differences, sorted recursive differences, shared multiplier based voltage scaling operation, architecture driven voltage scaling, power reduction using operation reduction and operation substitution. (9 Hours)
- **Multiple and Dynamic supply voltage design:** Multiple supply voltage design, dynamic supply voltage design, rate of change of supply voltages, power supply network, variation of the clock speed. (6 Hours)
- **Low power multi-core architectures:** Notion of multi-cores, hardware and software techniques for power reduction in multi-core architectures. (3 Hours)

Laboratory/practical/tutorial Modules:

- None

Textbooks:

1. K.S. Yeo, K. Roy, **Low Voltage Low Power VLSI Systems**, McGraw Hill, 2013.
2. A. Pal, **Low power VLSI design**, McGraw Hill, 2014.



References:

1. J. M. Rabaey and M. Pedram, Eds., **Low Power Design Methodologies**, Kluwer Academic Publishers, 1996.
2. B. Parhami, **Computer Arithmetic: Algorithms and Hardware Designs**, 2nd Edition, Oxford University Press, 2010.
3. I. Koren, **Computer Arithmetic Algorithms**, CRC Press, 2001.